CO 3 - ASSESSMENT TOOL 2

Data Interpretation - Questions and Step-by-Step Solutions

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1. Data Summarization

Question

A dataset of 11 employee salaries (in Rs. '000) is: 32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120.

(a) Compute the mean and median. (b) Explain why the mean is a poor measure of central tendency here. (c) Which measure of dispersion would you trust more - standard deviation or IQR - and why?

Solution

Step 1: Calculate the mean.

Mean = Sum of all salaries / Number of salaries

Sum = 32 + 35 + 36 + 38 + 40 + 42 + 45 + 48 + 50 + 95 + 120 = 581

n = 11

Mean = 581 / 11 = 52.82 (approximately) thousand rupees

Step 2: Calculate the median.

The data are already arranged in ascending order. Since n = 11 is odd, the median is the (11 + 1)/2 = 6th value.

Median = 42 thousand rupees

Step 3: Why is the mean poor here?

The salaries 95 and 120 are very large compared with most of the observations. These extreme high values pull the mean upward to 52.82, while the median remains 42. Therefore, the mean does not represent the typical employee salary very well.

Step 4: Which dispersion measure is more trustworthy?

IQR is more trustworthy because it is based on the middle 50% of observations and is resistant to outliers. Standard deviation is affected strongly by the extreme values 95 and 120.

Final Answer: Mean = Rs. 52.82 thousand; Median = Rs. 42 thousand; IQR is preferred over standard deviation for dispersion because the dataset contains strong high-end outliers.

2. Outlier Treatment

Question

Using the dataset above, apply the 1.5xIQR rule to identify outliers. Show Q1, Q3, IQR, and the lower/upper fences. Which values qualify as outliers? If you cap them at the fences (Winsorization) instead of removing them, how would the mean change qualitatively?

Solution

Step 1: Find Q1 and Q3.

There are 11 observations. The median is 42. Excluding the median, the lower half is 32, 35, 36, 38, 40 and the upper half is 45, 48, 50, 95, 120.

Q1 = middle value of lower half = 36

Q3 = middle value of upper half = 50

Step 2: Calculate the IQR.

IQR = Q3 - Q1 = 50 - 36 = 14

Step 3: Calculate the fences using the 1.5xIQR rule.

Lower fence = Q1 - 1.5(IQR) = 36 - 1.5(14) = 36 - 21 = 15

Upper fence = Q3 + 1.5(IQR) = 50 + 1.5(14) = 50 + 21 = 71

Step 4: Identify outliers.

Any observation below 15 or above 71 is an outlier. Therefore, 95 and 120 are outliers. The remaining values are within the fences.

Step 5: Winsorization and effect on the mean.

If the outliers are capped at the upper fence, 95 becomes 71 and 120 becomes 71. The new sum becomes 581 - 95 - 120 + 71 + 71 = 508.

New mean = 508 / 11 = 46.18 thousand rupees, approximately.

Thus, the mean decreases substantially because the two extreme salaries are pulled down to the upper fence, but the observations are not removed.

Final Answer: Q1 = 36, Q3 = 50, IQR = 14; Lower fence = 15; Upper fence = 71; Outliers = 95 and 120. After Winsorization, the mean falls from Rs. 52.82 thousand to about Rs. 46.18 thousand.

3. Measuring Asymmetry

Question

Two datasets have the same mean (50) and standard deviation (10), but Dataset A has Pearson's skewness coefficient = +1.2 and Dataset B has -0.8. Sketch (describe) the shape of each distribution and state which is more likely to have salary-like data with a long right tail.

Solution

Dataset A: Pearson skewness = +1.2.

A positive skewness means the distribution is right-skewed. Most observations are concentrated toward the lower or central side, while a longer tail extends to the right because of relatively large values.

Shape description: /___or a peak toward the left with a stretched tail toward the right.

Dataset B: Pearson skewness = -0.8.

A negative skewness means the distribution is left-skewed. Most observations are concentrated toward the higher side, while a longer tail extends to the left because of relatively small values.

Shape description: __/\ or a peak toward the right with a stretched tail toward the left.

Salary-like data often contain a small number of very high salaries compared with the majority of employees. That produces a long right tail and therefore positive skewness.

Final Answer: Dataset A is positively/right-skewed; Dataset B is negatively/left-skewed. Dataset A is more likely to represent salary data with a long right tail.

4. Coefficient of Skewness Calculation

Question

For a dataset: Mean = 62, Median = 58, Mode = 52, SD = 12. Calculate skewness using (a) Karl Pearson's coefficient (mode method), (b) Karl Pearson's coefficient (median method, Sk = 3(Mean - Median)/SD). Compare the two results and comment on why they might differ slightly.

Solution

(a) Karl Pearson's coefficient - mode method

Formula: Sk = (Mean - Mode) / SD

Substitute the values: Sk = (62 - 52) / 12 = 10 / 12 = 0.8333

Therefore, skewness by the mode method = +0.83 approximately.

(b) Karl Pearson's coefficient - median method

Formula: Sk = 3(Mean - Median) / SD

Substitute the values: Sk = 3(62 - 58) / 12 = 3(4) / 12 = 12 / 12 = 1.00

Therefore, skewness by the median method = +1.00.

Comparison: Both values are positive, so both methods indicate right/positive skewness. The values are 0.83 and 1.00, which are close but not identical.

Reason for the difference: The mode method uses the mode, while the median method uses the median. In real datasets the exact relationship among mean, median, and mode is not perfectly symmetric, so the two approximate Pearson measures can give slightly different results.

Final Answer: (a) Sk(mode) = +0.83 approximately. (b) Sk(median) = +1.00. Both indicate positive/right skewness; the small difference occurs because the two formulas use different central-tendency measures.

5. Variance vs Standard Deviation Interpretation

Question

Two production lines report these stats:

Line A: Mean = 200 units/day, Variance = 64

Line B: Mean = 150 units/day, Variance = 36

Which line has greater relative variability? Use the Coefficient of Variation to justify your answer, not just raw variance.

Solution

Step 1: Convert variance to standard deviation.

Standard deviation = sqrt(Variance).

Line A: SD = sqrt(64) = 8 units/day

Line B: SD = sqrt(36) = 6 units/day

Step 2: Calculate the Coefficient of Variation (CV).

Formula: CV = (SD / Mean) x 100%

Line A: CV = (8 / 200) x 100 = 4%

Line B: CV = (6 / 150) x 100 = 4%

Step 3: Compare the relative variability.

Although Line A has a larger raw variance (64 vs 36) and a larger standard deviation (8 vs 6), its mean production is also larger. After adjusting for the mean using CV, both lines have exactly the same relative variability of 4%.

Final Answer: Line A CV = 4%; Line B CV = 4%. Therefore, neither line has greater relative variability. Both production lines have equal relative variability.